



FINELITE

AI-powered credit card limit
prediction and risk analytics app

1. Executive Summary

The objective of this project is to **predict an appropriate customer Credit Limit** using demographic, financial, credit-history, spending, and repayment-related information.

Four regression models were evaluated:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- XGBoost Regressor

The analysis shows that **XGBoost is the strongest-performing model** among the models tested.

Key Findings

Model	Test R ²	Test MAE	Test RMSE	15-Fold CV Mean R ²
Linear Regression	96.47%	₹39,342	₹54,819	96.64%
Decision Tree	90.32%	₹71,991	₹90,782	91.95%
Random Forest	97.11%	₹39,478	₹49,605	97.47%
XGBoost	98.58%	₹27,772	₹34,776	98.86%

After hyperparameter tuning, XGBoost achieved a **5-fold cross-validation R² of 99.56%**, making it the preferred model for further evaluation and potential deployment.

2. Business Objective

The purpose of the project is to develop a machine-learning model that can estimate a customer's **Credit Limit** based on their financial profile and credit behavior.

A reliable Credit Limit prediction system can help the organization:

- Improve credit-limit assignment.
- Reduce manual decision-making.
- Support consistent credit decisions.
- Identify customers suitable for higher or lower limits.
- Improve customer segmentation.
- Reduce potential credit risk.
- Support relationship managers and credit teams with data-driven recommendations.

3. Model Development

Four regression algorithms were evaluated.

Model 1 — Linear Regression

Linear Regression was used as the baseline model.

It assumes that the relationship between the predictors and Credit Limit can be approximated using a linear equation.

Model 2 — Decision Tree

Decision Tree Regression was used to capture non-linear relationships.

Model 3 — Random Forest

Random Forest combines multiple decision trees to improve prediction stability and accuracy.

Model 4 — XGBoost

XGBoost is a gradient-boosting algorithm that builds trees sequentially and focuses on correcting errors made by previous trees.

It was expected to perform well because the relationship between customer financial characteristics and credit limit can be complex and non-linear.

4. Model Performance

4.1 Linear Regression

Test Results

R^2	=	96.47%
MAE	=	₹39,341.90
MSE	=	3,005,084,204.93
RMSE	=	₹54,818.65

Cross-Validation

15-fold cross-validation produced:

Mean R^2 = 96.64%

Interpretation

Linear Regression already provides a strong baseline, explaining approximately **96.47% of the variation** in Credit Limit on the test set.

However, it is outperformed by the ensemble models.

4.2. Decision Tree Results

Test Results

R^2 = 90.32%
MAE = ₹71,990.54
RMSE = ₹90,782.25

15-Fold Cross-Validation

Mean R^2 = 91.95%

The Decision Tree performed worse than Linear Regression, Random Forest, and XGBoost.

Tuned Decision Tree

GridSearchCV identified:

`criterion` = `squared_error`
`max_depth` = 15
`min_samples_leaf` = 4
`min_samples_split` = 10

Best 5-fold CV R^2 :

92.20%

Stakeholder Interpretation

A single decision tree does not provide the best predictive performance for this dataset.

4.3. Random Forest Results

Test Results

R^2 = 97.11%
MAE = ₹39,478.13
RMSE = ₹49,604.75

15-Fold Cross-Validation

Mean R^2 = 97.47%
Random Forest significantly improved over the single Decision Tree.

Tuned Random Forest

GridSearchCV selected:
n_estimators = 200
max_depth = None
min_samples_split = 2
min_samples_leaf = 1

Best 5-fold CV R^2 :
97.09%

Stakeholder Interpretation

Random Forest provides a strong and stable model and is a good alternative to XGBoost.

4.4. XGBoost Results

XGBoost produced the strongest initial performance.

Test Results

R² = 98.58%
MAE = ₹27,771.84
RMSE = ₹34,776.44

15-Fold Cross-Validation

Mean R² = 98.86%
This indicates excellent predictive consistency across different subsets of the dataset.

12. XGBoost Hyperparameter Tuning

GridSearchCV was used to identify better XGBoost parameters.
The tuning process evaluated:

n_estimators
learning_rate
max_depth
subsample
colsample_bytree

Best Parameters

n_estimators = 200
learning_rate = 0.1
max_depth = 3
subsample = 0.8
colsample_bytree = 1.0

Best 5-Fold Cross-Validation R²
99.56%

This is the strongest validation result obtained in the project.

5. Why XGBoost is the Recommended Model

XGBoost is recommended because it provides the best overall performance across the key evaluation metrics.

Compared with Linear Regression

XGBoost reduced MAE from approximately:

₹39,342 → ₹27,772

That represents an approximate **29% reduction in average absolute prediction error**.

Compared with Random Forest

XGBoost reduced MAE from:

₹39,478 → ₹27,772

and RMSE from:

₹49,605 → ₹34,776

Therefore, XGBoost provides both **higher predictive accuracy and lower prediction error**.

```
: from sklearn.model_selection import GridSearchCV
from xgboost import XGBRegressor
tunned_parameters_xgb = {'n_estimators': [100, 200],
                        'learning_rate': [0.05, 0.1],
                        'max_depth': [3, 5],
                        'subsample': [0.8, 1.0],
                        'colsample_bytree': [0.8, 1.0]}
model_gscv_xgb = XGBRegressor(objective='reg:squarederror', random_state=42, n_jobs=-1)
new_model_xgb = GridSearchCV(estimator=model_gscv_xgb, param_grid=tunned_parameters_xgb, scoring='r2', cv=5, n_jobs=-1)
new_model_xgb.fit(xtrain, ytrain)
print("Best Parameters:", new_model_xgb.best_params_)
print("\nBest R2 Score:", new_model_xgb.best_score_)

Best Parameters: {'colsample_bytree': 1.0, 'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 200, 'subsample': 0.8}

Best R2 Score: 0.9956006169319153
```

6. Business Interpretation of Model Performance

The tuned XGBoost model achieved:

99.56% R^2 during 5-fold cross-validation

This means the model explains approximately **99.56% of the variation in Credit Limit within the cross-validation process.**

However, this should **not** be interpreted as "99.56% accurate."

R^2 measures the proportion of variance explained by the model; it is not the percentage of predictions that are correct.

"The tuned XGBoost model demonstrated very strong predictive performance, achieving a cross-validated R^2 of 99.56%."